



Original research

Effects of the Elevation Training Mask® 2.0 on dyspnea and respiratory muscle mechanics, electromyography, and fatigue during exhaustive cycling in healthy humans

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ARTICLE INFO

Article history:

Received 11 June 2021

Received in revised form 27 August 2021

Accepted 29 August 2021

Available online 3 September 2021

Keywords:

Diaphragm

Exercise performance

Breathlessness

Respiratory loading

ABSTRACT

Objectives: Examine the effects of the Elevation Training Mask® 2.0 (ETM) on dyspnea, and respiratory muscle function and fatigue during exercise.

Design: Randomized crossover.

Methods: 10 healthy participants completed 2 time-to-exhaustion (TTE) cycling tests while wearing the ETM or under a sham control condition. During the sham, participants were told they were breathing air equivalent to “9000 ft” (matched to the selected resistance valves on the ETM according to the manufacturer), but they were breathing room air. Dyspnea and leg discomfort were assessed using the modified 0–10 category-ratio Borg scale. Qualitative dyspnea descriptors at peak exercise were selected from a list of 15. Crural diaphragmatic electromyography (EMG_{di}) and transdiaphragmatic pressure (P_{di}) were measured via a multipair esophageal electrode balloon catheter. Participants performed maximal respiratory maneuvers before and after exercise to estimate the degree of respiratory muscle fatigue.

Results: Exercise with the ETM resulted in a significant decrease in TTE ($p = 0.015$), as well as increased dyspnea at baseline ($p = 0.032$) and during the highest equivalent submaximal exercise time ($p = 0.0001$). The increase in dyspnea with the ETM was significantly correlated with the decrease in exercise time ($r = 0.73$, $p = 0.020$). EMG_{di} and P_{di} were significantly increased with the ETM at all time points (all $p < 0.05$). There was a significant increase in the selection frequency of “my breath does not go in all the way” at peak exercise with the ETM ($p = 0.02$). The ETM did not induce respiratory muscle fatigue.

Conclusions: Exercising with the ETM appears to decrease exercise performance, in part, by increasing the sensation of dyspnea.

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Practical implications

- Individuals may want to avoid using the ETM during high-intensity exercise training due to its reduction in exercise performance, which is partly caused by increases in dyspnea.
- The ETM increases diaphragmatic EMG and pressure generation compared to exercise without the ETM, which could provide a sufficient training stimulus to the respiratory muscles.
- The ETM may be most beneficial as a respiratory muscle training device at low intensity exercise such as warm-up, cool-downs and recovery exercise.

1. Introduction

The Elevation Training Mask® 2.0 (ETM) was originally purported to simulate high altitude and has been growing in popularity. The ETM causes modest decreases in oxygen saturation (S_pO_2) during exercise in some,^{1–4} but not all studies.⁵ In addition, there are no hematological changes in healthy participants following 6-week cycle training with the ETM.⁶ Therefore, any hypoxemia with the ETM is not caused by simulating altitude, but likely due to its modest dead space and impairment of alveolar ventilation as previously suggested.^{2,4} It is likely that these findings resulted in a shift in marketing from an altitude to a respiratory muscle loading device.

Several weeks of isolated respiratory muscle training can improve exercise performance.^{7,8} However, increasing the respiratory load simultaneously during endurance training may actually decrease overall exercise performance compared to endurance training without added

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respiratory resistance. This could happen because of the increase in transdiaphragmatic pressure (P_{di}) and potential for respiratory muscle fatigue, which might reduce overall training load on the locomotor muscles by decreasing exercise time and/or intensity.^{9,10} However, this hypothesis remains speculative as no studies have evaluated P_{di} and respiratory muscle fatigue with the ETM.

Increases in dyspnea may also reduce exercise time/intensity and thus subsequent training load on the locomotor muscles during combined respiratory loading and whole-body exercise. This hypothesis is based on the link between increased dyspnea and decreased performance.¹¹ Emerging evidence suggests that exertional dyspnea is primarily caused by an increased awareness of inspiratory neural drive (IND), estimated using diaphragm electromyography (EMG_{di}).¹² Wearing the ETM during acute bouts of exercise has been shown to increase dyspnea⁴; however, the qualitative descriptors of dyspnea, and whether EMG_{di} also increases remains unknown. Accordingly, the purpose of this study was to determine the effect of the ETM on exercise time-to-exhaustion (TTE), dyspnea, diaphragmatic function, and respiratory muscle fatigue in healthy humans. We hypothesized that the ETM would decrease TTE and increase dyspnea, EMG_{di} , P_{di} , and respiratory muscle fatigue compared to control exercise.

2. Methods

This study received ethical approval (H16-02589) with participants providing written informed consent and took place over 3 visits separated by at least 48 h. Participants were instructed to avoid caffeine and exercise prior to each visit. Visit 1 included spirometry¹³ until both forced vital capacity (FVC) and forced expiratory volume in 1-second (FEV_1) were reproducible within 0.15 l. Participants then underwent an incremental cycling test (Velotron, Racermate, Seattle, WA, USA) starting at 50 W for females and 100 W for males, with 25 W stepwise increases every 2 min until exhaustion. Following 30 min of rest, participants underwent a familiarization TTE constant load cycling test at 85% of maximum incremental work rate, which is an intensity known to cause respiratory muscle fatigue.¹⁴ TTE tests are reproducible even in naive participants,¹⁵ and therefore only one familiarization test was performed to diminish the learning effect. Heart rate (HR) (Polar H10, Polar Electro, Kempele, Finland) and ventilatory/metabolic data (TrueOne 2400, Parvo Medics, Sandy, UT, USA) were recorded during both exercise tests.

During the subsequent visits, participants repeated the TTE test under two randomized conditions: 1) ETM with the resistance valves set to mimic “9000 ft” according to the manufacturer; and 2) a sham condition where participants were told they were breathing a gas mixture equivalent to 9000 ft, but were actually breathing room air. The deception involved participants breathing through a mouthpiece connected to a two-way-non-rebreathing valve in which the inspiratory port appeared to be connected to a filled 50 l Douglas bag. Participants were led to believe that we were testing the effectiveness of the ETM to simulate altitude and that the control condition was the “gold standard”. The sham served as a control and to minimize the placebo effect of the ETM. To avoid participants becoming unblinded, we only recruited individuals that were naive to exercise physiology experiments. Participants were unblinded at the end of the study, and no participants indicated they were aware of the deception.

Crural EMG_{di} was recorded via a multipair esophageal electrode catheter (Yinghui Medical Technology Co. LTD, Guangzhou, China). The catheter itself is 2.8 mm in diameter and consists of 9 1 cm silver electrodes that make up 5 recording pairs distanced 3.4 cm apart. The catheter was inserted through the nose following numbing with a topical anesthetic and positioned to optimize the EMG_{di} signals. Catheter position was secured in place when the EMG amplitude in the middle recording pair was lowest and highest in the two furthest outer pairs during spontaneous breathing.¹⁶ Raw EMG_{di} signals were amplified and band pass filtered between 20 and 1000 Hz (bio-amplifier model

RA-8, Yinghui Medical Technology Co. LTD, Guangzhou, China) and converted to root mean square using a 0.1 s moving average window. Values were selected on a breath-by-breath basis between QRS complexes. The value selected for each breath was the highest value recorded in the 5 electrode pairs. EMG_{di} was expressed as a percentage of the highest value obtained during inspiratory capacity maneuvers on a given visit. The same catheter was used and placed in the same position within the same nostril on both visits.

The catheter used to measure EMG_{di} included two balloons connected to differential pressure transducers (DP15-34, Validyne Engineering, Northridge, CA, USA) to measure gastric (P_{ga}) and esophageal (P_{es}) pressures. Esophageal and gastric balloons were filled with 0.5 and 1.2 ml of air, respectively. P_{di} was calculated as the difference between P_{ga} and P_{es} . Maximal P_{di} swings were quantified during inspiration while decreases in volitional pressure generation following exercise was used to determine the presence of respiratory muscle fatigue. Briefly, participants performed maximal sniffs, maximal inspiratory pressure (MIP) maneuvers at residual volume, and maximal expiratory pressure (MEP) maneuvers at total lung capacity¹⁷ before, immediately after, and 15 min after exercise in an upright seated position. A decrease in peak P_{di} during a sniff or the highest 1-second mean P_{di} and mouth pressure (P_m) during a MIP maneuver was used to investigate inspiratory muscle fatigue, whereas a decrease in the highest 1-second mean P_{ga} and P_m during a MEP maneuver was used to investigate expiratory muscle fatigue. All maneuvers were repeated 3–10 times, with the average of the highest three maneuvers within 10% being used in the analysis. The threshold for fatigue was either a decrease $\geq 15\%$ or double the coefficient of variation if it was $\geq 15\%$ for a given maneuver during the same visit. Finally, the tension time index of the diaphragm (TTI_{di}) was calculated by multiplying the ratio between mean inspiratory P_{di} and the maximum P_{di} achieved during a MIP maneuver performed on the same visit by the duty cycle.

Cardiorespiratory data could not be collected during the ETM condition. However, breathing frequency (f_b) was quantified by counting EMG_{di} bursts. HR was recorded as in visit 1 and SpO_2 was measured via finger pulse oximetry (Radical-7, Masimo Corporation, Irvine, CA, USA). During exercise, dyspnea and leg discomfort were assessed every minute. Dyspnea was defined as “the sensation of laboured or difficult breathing” and leg discomfort was defined as “the sensation of fatigue in the leg muscles.” Both sensations were evaluated using the modified 0–10 category-ratio Borg scale¹⁸ in which 0 represented “no breathing/leg discomfort” and 10 represented “the most breathing/leg discomfort ever experienced or imagined.” Participants were also asked every 2 min to select their most prominent dyspnea descriptor as either 1) “My breathing requires more work and effort”; 2) “I cannot get enough air in”; 3) “I cannot get enough air out”; or 4) “None apply”.^{19,20} Following exercise termination, participants then selected qualitative dyspnea descriptors experienced at peak exercise using a modified list of 15 from Simon et al.²¹ Lastly, participants selected their primary reason for stopping exercise as “breathing discomfort”, “leg discomfort”, “a combination of breathing and leg discomfort” or “other” and attributed a percentage towards stopping to either breathing or leg discomfort.

All signals were collected using a data acquisition system (PowerLab 16/35, ADInstruments, Colorado Springs, CO, USA) at 2000 Hz. Physiological variables were averaged over a 30-second time period in which no sensory responses were obtained every 2 min. Normality was tested using the Shapiro-Wilk test. A two-way repeated-measures ANOVA (condition and time) was used to detect differences in exercise responses at baseline; the highest equivalent submaximal exercise time (HESET) a participant achieved on both visits; and peak exercise, defined as the final 30 s of exercise. Bonferroni post hoc comparisons were made between conditions at the same time points. A two-way repeated-measures ANOVA (condition and time) was also used to detect differences in respiratory muscle fatigue, with Tukey's post hoc test used to detect differences between time points in the same

condition. Spearman's correlation coefficient was used to determine the relationship between changes in TTE and dyspnea, while a Pearson correlation coefficient was used to assess the relationship between changes in TTE and volitional respiratory muscle pressure generation from pre-exercise to immediately post exercise between conditions. Differences in TTE and the percentage contribution towards stopping exercise were assessed with paired *t*-tests. Differences between peak dyspnea descriptors were compared using McNemar's test. Statistical significance was set at $p < 0.05$. Values are mean $\pm$ SD unless otherwise stated.

3. Results

Fourteen participants were recruited but only 10 completed all visits and have data presented. Table 1 shows participant characteristics and peak exercise data. Participants displayed normal spirometry (i.e., FEV₁ > 80% predicted²² and FEV₁/FVC > 0.70).

There was a decrease in TTE with the ETM compared to sham (5.9 ± 3.0 vs. 9.5 ± 5.6 min, $p = 0.015$) (Fig. 1C). Table 2 shows TTE responses at baseline, HESET and peak. There was a significant effect of condition ($p = 0.0003$) and time ($p < 0.0001$) on EMG_{di}, which significantly increased with the ETM at baseline ($p = 0.001$), HESET ($p = 0.004$) and peak ($p = 0.010$). Similarly, P_{di} swings showed a main effect of condition ($p = 0.0005$) and time ($p < 0.0001$) and significantly increased with the ETM at all time points (baseline $p = 0.001$; HESET $p = 0.037$; peak $p = 0.042$). There was a significant main effect of condition and time ($p = 0.0024$ and $p < 0.0001$, respectively) on f_b with a significant decrease with the ETM at baseline only ($p = 0.015$). There was no effect of condition or time on SpO₂ (both $p > 0.05$) and an effect of time only ($p < 0.0001$) for HR.

Perceptual responses are shown in Fig. 1. There was a significant effect of condition ($p = 0.0002$) and time ($p < 0.0001$) on Borg dyspnea ratings with an increase in dyspnea by an average of 0.4 ± 0.4 and 4.0 ± 1.7 units at baseline ($p = 0.032$) and HESET ($p = 0.0001$), respectively, but no

significant difference at peak ($p = 0.082$). The mean change in dyspnea across the three analysis time points between conditions was significantly correlated with the changes in TTE ($r = 0.73$, $p = 0.020$). There was a significant main effect of time ($p < 0.0001$) but not condition ($p = 0.67$) on leg discomfort. At HESET, 5 additional participants selected “I cannot get enough air in” as their primary dyspnea descriptor compared to the control condition, while 3 additional participants selected this descriptor with the ETM at peak. At peak exercise, “my breath does not go in all the way” was the only dyspnea descriptor that showed significant differences in selection frequency ($p = 0.02$) (Fig. 1A) although “I feel that I am suffocating” and “I cannot take a deep breath in” approached statistical significance (both $p = 0.06$). During the sham trial, no participants selected “breathing discomfort” as their primary reason for stopping exercise, while 6 participants selected it as their primary reason with the ETM. The percentage contribution between “breathing discomfort” and “leg discomfort” as the primary reason for stopping exercise was $37 \pm 21\%$ and $63 \pm 21\%$, respectively, in the sham trial ($p = 0.004$), which was nearly reversed with the ETM (“breathing discomfort” = $67 \pm 22\%$, “leg discomfort” = $33 \pm 22\%$, $p = 0.004$).

There was no effect of time nor condition on P_{di} during maximal sniffs or P_{di} and P_m during MIP maneuvers (all $p > 0.05$). There was an effect of time only on both P_{ga} ($p = 0.028$) and P_m ($p = 0.027$) during MEP maneuvers. Post hoc comparisons revealed there were no significant differences between any time points in both conditions (all $p > 0.05$). Individually, 2 participants met our definition of fatigue in P_{di} during a maximal sniff in the sham condition compared to 3 with the ETM. During MIP maneuvers, 2 participants showed fatigue in P_{di} in both conditions. Similarly, fatigue was observed in P_m in 2 participants in the sham condition compared to 1 with the ETM. During MEP maneuvers, 2 and 4 participants showed fatigue in P_{ga} and P_m, respectively, in both conditions. There were no significant correlations between the change in TTE and change in pre/post exercise volitional pressure generation between conditions with any maneuver (sniff: P_{di} $r = -0.17$; MIP: P_{di} $r = -0.32$, P_m $r = -0.18$; MEP: P_{ga} $r = 0.08$, P_m $r = 0.20$; all $p > 0.05$). There was a significant effect of time ($p < 0.0001$) and condition ($p = 0.0008$) on TTI_{di}. Post hoc tests revealed TTI_{di} was significantly increased with the ETM at all time points (baseline $p = 0.005$; HESET $p = 0.005$; peak $p = 0.016$). Individually, 2 participants had a TTI_{di} > 0.15 at both HESET and peak in the sham trial, while 5 participants did at the same time points during exercise with the ETM.

4. Discussion

The ETM was shown to: 1) increase dyspnea intensity ratings and the selection frequency of “I cannot get enough air in”; 2) reduce TTE, which was significantly correlated with the increase in dyspnea ratings; 3) increase EMG_{di} and P_{di}; and 4) not increase the occurrence of respiratory muscle fatigue compared to control.

The ETM has been shown to reduce exercise performance during both incremental¹ and constant load exercise.⁴ The present study demonstrated a mean reduction in TTE of $33 \pm 26\%$. An increase in dyspnea is one mechanism by which the ETM could reduce TTE. Both the present study and the work of others⁴ showed a significant increase in dyspnea ratings during submaximal exercise with the ETM. We also showed a strong correlation between the change in dyspnea and TTE between ETM and sham. The increase in dyspnea with the ETM is likely due, at least in part, to the significant increase in IND, as reflected by the increase in EMG_{di}. The increase in IND appears to have been caused by the substantial increase in P_{di} needed to overcome the increased resistance of the ETM compared to the sham. IND may also increase due to modest hypoxemia caused by the increased dead space of the ETM, which has been shown by some^{1–4} but not others.⁵ However, SpO₂ did not change between conditions in the present study and is therefore an unlikely contributor to the increased IND. It is also possible that dyspnea may have increased due, in part, to a small dead space-induced increase in minute ventilation ($\dot{V}_E$), which could not be

Table 1
Participant characteristics and peak exercise responses.

<i>Demographics/Anthropometrics</i>	
Sex, M:F	5:5
Age, years	28 $\pm$ 5
Height, cm	174.2 $\pm$ 8.5
Mass, kg	74.7 $\pm$ 9.6
BMI, kg $\cdot$ m ⁻²	24.5 $\pm$ 1.9
<i>Spirometry</i>	
FVC, l	5.34 $\pm$ 1.03
FVC, % predicted	108 $\pm$ 14
FEV ₁ , l	4.36 $\pm$ 0.94
FEV ₁ , % predicted	104 $\pm$ 9
FEV ₁ /FVC, %	79 $\pm$ 6
<i>Peak incremental exercise</i>	
Work Rate, W	298 $\pm$ 55
$\dot{V}O_2$, l $\cdot$ min ⁻¹	3.45 $\pm$ 0.94
$\dot{V}O_2$, ml $\cdot$ kg ⁻¹ $\cdot$ min ⁻¹	46.0 $\pm$ 9.6
$\dot{V}CO_2$, l $\cdot$ min ⁻¹	3.79 $\pm$ 1.04
RER	1.10 $\pm$ 0.08
HR, beats $\cdot$ min ⁻¹	178 $\pm$ 14
$\dot{V}_E$, l $\cdot$ min ⁻¹	131.8 $\pm$ 41.9
$\dot{V}_T$, l	2.71 $\pm$ 0.62
f_b , breaths $\cdot$ min ⁻¹	47.7 $\pm$ 8.5
$\dot{V}_E/\dot{V}O_2$	37.6 $\pm$ 5.4
$\dot{V}_E/\dot{V}CO_2$	34.2 $\pm$ 3.2
Dyspnea, 0–10 Borg scale	7.2 $\pm$ 2.5
Leg discomfort, 0–10 Borg scale	8.5 $\pm$ 1.4

Spirometry data are from 7 participants due to technical issues with 3 participants. Abbreviations: M, male; F, female; BMI, body mass index; FVC, forced vital capacity; FEV₁, forced expiratory volume in 1 s; $\dot{V}O_2$, oxygen consumption; $\dot{V}CO_2$, carbon dioxide production; RER, respiratory exchange ratio; HR, heart rate; $\dot{V}_E$, minute ventilation; $\dot{V}_T$, tidal volume; f_b , breathing frequency; $\dot{V}_E/\dot{V}O_2$, ventilatory equivalent for oxygen; $\dot{V}_E/\dot{V}CO_2$, ventilatory equivalent for carbon dioxide. Values are mean $\pm$ SD.

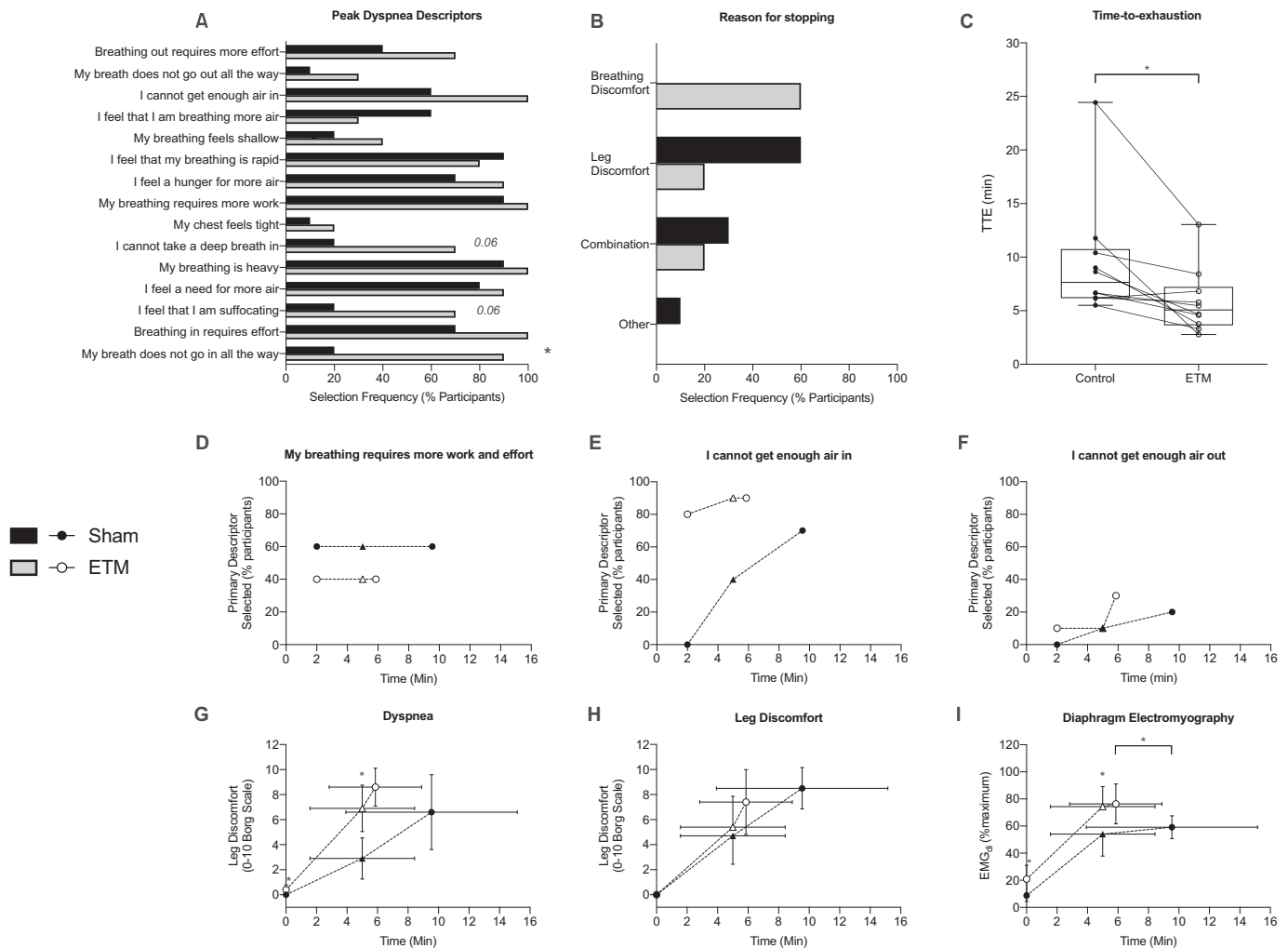


Fig. 1. A: selection frequency of 15 dyspnea descriptors post exercise. B: reasons for stopping exercise. C: box and whisker plot and individual data points for time-to-exhaustion tests. Solid line in box and whisker plots represents median value, while whiskers represent minimum and maximum exercise times. Dashed lines connect individual participant data between conditions. D: selection frequency of “my breathing requires more work and effort”. E: selection frequency of “I cannot get enough air in”. F: selection frequency of “I cannot get enough air out”. G: dyspnea intensity ratings over time. H: leg discomfort ratings over time. I: diaphragmatic EMG over time. Values on panels D–F are plotted across measurement time points all participants achieved [2nd minute, highest equivalent submaximal exercise time (HESET) and peak]. Values on panels G, H and I are plotted at analysis time points [baseline (0 min), HESET and peak]. Triangle represents HESET in panels D–I. Panels A, B, D–F are percent of participants, while panels G–I are mean $\pm$ SD. *, $p < 0.05$ statistically significant between conditions.

measured in the ETM condition in this study. However, this is unlikely given that breathing frequency was actually lower with the ETM during exercise, and previous work shows a decrease rather than an increase in $\dot{V}_E$ with the ETM.^{2,5} Nevertheless, we cannot draw definitive conclusions regarding $\dot{V}_E$ since we were unable to measure $\dot{V}_T$ in this study. Another potential mechanism of increased dyspnea is dynamic hyperinflation; however, we were unable to determine if hyperinflation occurred in our subjects since we could not measure inspiratory capacity in the ETM condition. Finally, we must acknowledge that increases in facial skin temperature, temperature of the inhaled air, and increased humidity could potentially contribute to higher dyspnea with the ETM, although this requires further investigation.²³

The ETM could also limit TTE due to increased respiratory muscle contractions and respiratory muscle fatigue. Two possible mechanisms for this are 1) the influence of respiratory muscle fatigue on dyspnea¹¹ and 2) the respiratory muscle metaboreflex, a sympathetically mediated reflex that diverts blood away from the locomotor muscles following increased respiratory muscle work to the point of fatigue.^{24,25} There was an effect of time on maximal expiratory, but not inspiratory pressure generation. However, post hoc tests revealed no significant differences in maximal expiratory pressures between time points in both conditions. Thus, on average, there was no effect of the ETM on respiratory muscle

fatigue. There were a handful of participants that met our definition of fatigue but this did not differ appreciably from the sham condition. Thus, it is unlikely that the increased dyspnea with the ETM was related to respiratory muscle fatigue. Despite the lack of evidence of respiratory muscle fatigue, it is still possible that the ETM could have triggered the respiratory metaboreflex and could therefore have contributed to the reduced TTE. However, the lack of change in leg discomfort ratings and HR indirectly suggest that the metaboreflex was likely not the primary contributor to reduced exercise performance in this study, despite the difference in TTI_{di} between conditions. Collectively, the findings of this study suggests that dyspnea, and not increased respiratory muscle fatigue or an increase in the respiratory metaboreflex, is probably the primary mechanism by which the ETM limits exercise performance. This suggestion is further supported by a greater selection frequency of multiple distressing qualitative descriptors of dyspnea and that 80% of our participant's selected breathing discomfort alone or in combination with leg discomfort as the primary reason for stopping exercise with the ETM compared to only 30% in the sham condition.

This study reinforces the fact that the ETM is a respiratory muscle loading device. Indeed, baseline P_{di} swings with the ETM were 37% of the maximum P_{di} achieved during a pre-exercise MIP maneuver, 65% at HESET, and 63% at peak, respectively. Many inspiratory muscle training

Table 2

Exercise responses and volitional pressure generation with the ETM.

Cardiorespiratory and sensory measures	Baseline		HESET		Peak exercise	
	Sham	ETM	Sham	ETM	Sham	ETM
EMG _{di} , % of maximum	8.7 ± 4.3	20.3 ± 10.8*	54.1 ± 16.2	75.5 ± 13.7*	59.2 ± 8.5	78.9 ± 11.6*
HR, beats·min ⁻¹	61 ± 7	65 ± 7	159 ± 15	162 ± 14	173 ± 8	169 ± 12
f _b , breaths·min ⁻¹	15.5 ± 3.3	11.9 ± 3.4*	36.4 ± 7.0	35.2 ± 8.1	47.4 ± 6.9	42.4 ± 7.6
P _{di} swings, cmH ₂ O	8.2 ± 3.4	19.2 ± 6.4*	26.1 ± 11.0	34.6 ± 13.1*	26.4 ± 7.6	33.5 ± 9.4*
TTI _{di}	0.03 ± 0.02	0.07 ± 0.04*	0.09 ± 0.05	0.15 ± 0.06*	0.10 ± 0.04	0.16 ± 0.07*
SpO ₂ , %	99 ± 1	99 ± 2	98 ± 2	97 ± 2	97 ± 3	97 ± 3
Dyspnea, 0–10 Borg scale	0.0 ± 0.0	0.4 ± 0.4*	2.9 ± 1.6	6.9 ± 1.9*	6.6 ± 3.0	8.6 ± 1.5
Leg Discomfort, 0–10 Borg scale	0.0 ± 0.0	0.0 ± 0.0	4.7 ± 2.2	5.4 ± 2.5	8.5 ± 1.7	7.4 ± 2.6
Volitional pressure generation	Pre-exercise		Post-exercise		15-min post-exercise	
	Sham	ETM	Sham	ETM	Sham	ETM
Sniff P _{di} , cmH ₂ O	105.8 ± 28.8	101.9 ± 17.9	99.8 ± 16.8	97.2 ± 24.0	109.5 ± 23.8	100.9 ± 16.5
MIP P _{di} , cmH ₂ O	73.0 ± 38.3	65.7 ± 32.1	70.2 ± 19.7	68.1 ± 16.3	73.7 ± 28.9	68.0 ± 20.4
MIP P _m , cmH ₂ O	135.2 ± 23.3	135.4 ± 23.7	128.1 ± 25.2	131.4 ± 19.6	130.5 ± 24.0	132.8 ± 24.0
MEP P _{ga} , cmH ₂ O	126.1 ± 40.4	131.5 ± 23.6	116.6 ± 39.9	121.3 ± 34.3	115.8 ± 38.5	123.2 ± 28.7
MEP P _m , cmH ₂ O	165.8 ± 34.1	162.6 ± 37.4	153.7 ± 37.0	148.9 ± 34.2	152.5 ± 36.1	158.7 ± 31.5

Abbreviations: EMG_{di}, diaphragm electromyography; HR, heart rate; f_b, breathing frequency; P_{di}, transdiaphragmatic pressure; P_{es}, esophageal pressure; P_{ga}, gastric pressure; SpO₂, peripheral oxygen saturation. *n* = 9 for Sniff P_{di}, MIP P_{di} and MEP P_{ga} due to catheter displacement at exercise cessation in one participant. Values are mean ± SD.

* Significantly different between conditions at the same time point (*p* < 0.05).

programs are performed at 30–60% of MIP, which can increase respiratory muscle strength.²⁶ Therefore, it seems reasonable to speculate that the ETM could provide a sufficient respiratory muscle training load to improve respiratory muscle strength. Increasing the strength of the respiratory muscles can improve whole-body exercise performance.^{26,27} However, given that the ETM acutely decreases performance as shown by us and others,^{1,4} we believe that using the ETM may be counterproductive, at least during high-intensity whole-body exercise training where overall training intensity should not be compromised. For instance, athletes may need to reduce the exercise intensity to complete a given training session in order to compensate for the increased respiratory load and dyspnea. It is possible the ETM may be better suited during lower intensity exercise sessions (e.g. warm-up, cool-downs, recovery workouts, or even during resistance training²⁸). This approach would allow individuals to gain the benefits of training the respiratory muscles without having to compromise the intensity and quality of their high-intensity whole-body training sessions. However, we acknowledge that this remains speculative since the incorporation of the ETM in various exercise training programs still needs to be further explored.

There are several limitations of this study. First, the use of volitional pressure generation maneuvers to assess fatigue has a number of drawbacks including the reliance on participant motivation. For this reason, the criteria for variability allowed between maneuvers was only 10%. A non-volitional method such as supramaximal phrenic nerve stimulation would remove motivation as a potential confounder and should be further explored. Second, the exercise intensity used does not reflect the effect of the ETM during light or moderate intensity exercise. Third, we only tested one resistance setting and one ETM model, and therefore our findings cannot be extrapolated to the other resistance settings or masks (e.g. ETM 3.0). Fourth, there is no optimal control condition when testing the ETM. Lastly, given that the catheter was positioned to optimize EMG_{di}, the P_{es} balloon may not have been in the optimal position in some participants. However, the same catheter was always placed in the same position between visits. Future studies are needed to validate P_{es} when assessed using a catheter optimized to measure EMG via the occlusion technique.

5. Conclusion

Exercising with the ETM limits exercise performance due, in part, to an increase in the intensity of dyspnea and not an increase in respiratory

muscle fatigue. The ETM also significantly increases the selection frequency of unpleasant qualitative descriptors of dyspnea along with increases in EMG_{di} and P_{di} during exercise. Future studies are needed to determine the most appropriate training sessions to use the ETM in order to maximize the benefits of the device without compromising the quality of individual whole-body exercise training sessions.

Funding information

This study was funded by a Discovery Grant from the Natural Sciences and Engineering Research Council (NSERC) of Canada. Gabby Napoleone was supported by an Undergraduate Student Research Award from the NSERC. Andrew Ramsook was supported by a NSERC Canada Graduate Scholarship. Reid A. Mitchell was supported by a Postgraduate Scholarship from the NSERC and a University of British Columbia 4-year fellowship. Jordan A. Guenette was supported by a Scholar Award from the Michael Smith Foundation for Health Research and a Canadian Institutes of Health Research Clinical Rehabilitation New Investigator Award.

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Confirmation of ethical compliance

The research study conformed to the Declaration of Helsinki, excluding registration of a database, and received ethical approval from The University of British Columbia and Providence Health Care Research Institute Ethics Board (UBC-PHC REB Number: H16-02589). The research in this manuscript has not been published nor submitted elsewhere, nor will it be submitted elsewhere until a final decision from the Journal of Science and Medicine in Sport is made.

Acknowledgments

We thank our research participants for enthusiastically volunteering for this study.

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